

MATH 2318 – Linear Algebra**Midterm Review: Topics and Examples**

Midterm: Tuesday, October 13, in class.

References are to Lay, Lay & McDonald, *Linear Algebra and Its Applications*, 6th ed.

Topic	Section	Examples
<i>Chapter 1</i>		
Row reduction to echelon and reduced echelon form; pivot positions	1.2	2, 3
Basic and free variables; general solution; existence and uniqueness	1.2	4, 5
Solution sets of homogeneous systems; parametric vector form	1.5	1, 2
Solution sets of nonhomogeneous systems and their relation to the homogeneous system	1.5	3
The standard matrix of a linear transformation	1.9	1, 2, 5
<i>Chapter 2</i>		
Inverse of a 2×2 matrix	2.2	1, 2
Finding A^{-1} by row reducing $\begin{bmatrix} A & I \end{bmatrix}$	2.2	7
The Invertible Matrix Theorem; invertible linear transformations	2.3	1, 2
Subspaces of $\mathbb{R}^n$	2.8	1, 2, 3
Null space: basis for $\text{Nul } A$	2.8	6
Column space: basis for $\text{Col } A$	2.8	7, 8
Dimension and rank; the Rank Theorem	2.9	2, 3
<i>Chapter 3</i>		
Determinants by cofactor expansion	3.1	1, 2
Determinants of triangular and nearly triangular matrices	3.1	3

Also review the assigned homework problems from these sections and Quizzes 1–4.